Evidence is not the same as an assumption.

Every site-specific engineering and commercial input has an explicit range. Full register: model/assumptions.csv and inputs.json.

Notation: central value [low–high]. Ranges are plausible screening bounds, not confidence intervals. All proposed customer demand, capture capacity, routes, CAPEX and contracts remain unvalidated. Source IDs resolve on A4–A5; p means PDF page, s means slide.

Input / registry ID	Central [range]	Basis and validation
Urban recoverable tranche U_source_mw	2.5 [1.5–3.5] MW recovered available atinterface	A · generic context S15 p42; verify circuit telemetry and rights.
Urban capture-water inlet U_source_c	30 [24–35] degC	A · generic context S15 p42; verify circuit telemetry and rights.
Urban assumed source drop U_source_dt	8 [5–12] K	A · generic context S15 p42; verify circuit telemetry and rights.
Rural capture-water inlet R_source_c	45 [35–50] degC	A · generic context S15 p42; verify circuit telemetry and rights.
HP nominal heating / urban U_hp_mw	3.0 [1.5–4] MW heating	A · design screen; verify in detailed engineering. Full source/range register supplied.
Network supply P_supply	65 [63–70] degC	A · design screen; verify in detailed engineering. Full source/range register supplied.
Network design return P_return	40 [35–45] degC	A · design screen; verify in detailed engineering. Full source/range register supplied.
DHW production target P_dhw	60 [60–60] degC	A · design screen; verify in detailed engineering. Full source/range register supplied.
Exchanger approach P_hx_approach	3 [2–5] K	A · design screen; verify in detailed engineering. Full source/range register supplied.
Fraction of Carnot COP P_carnot_eta	0.52 [0.44–0.6] fraction	A · S17 s26,40; replace with certified vendor map.
Pipe conductance, per pipe P_u_pipe	0.4 [0.2–0.65] W/m/K per pipe	A · design screen; verify in detailed engineering. Full source/range register supplied.
Urban total trench U_route_m	800 [400–1400] m one-way trench	A · route allowance; survey branches and easements.
Rural total trench R_route_m	3500 [2000–5000] m one-way trench	A · route allowance; survey branches and easements.
Urban tank volume U_storage_m3	150 [75–250] m3	A · S11 storage context; test usable stratification.
Usable stratified volume P_storage_usable	0.85 [0.7–0.95] fraction geometric volume	A · S11 storage context; test usable stratification.
Source outage / urban U_source_outage_h	168 [72–336] h/year	A · design screen; verify in detailed engineering. Full source/range register supplied.

The assumptions that can reverse the recommendation.

Model inputs use real 2026 dollars and useful customer heat, unless the unit explicitly says otherwise.

Input / registry ID	Central [range]	Basis and validation
Target urban homes U_homes	900 [500–1200] target homes	A · prospective cohort; meter loads and secure subscriptions.
DHW per target home U_housing_dhw	2.5 [1.8–3.5] MWh/home/year useful	A · prospective cohort; meter loads and secure subscriptions.
Commercial DHW/process U_commercial_dhw	6500 [4000–8500] MWh/year useful	A · prospective cohort; meter loads and secure subscriptions.
Commercial space heat U_commercial_space	3500 [2000–5000] MWh/year useful	A · prospective cohort; meter loads and secure subscriptions.
Rural target homes R_homes	300 [200–500] target homes	A · prospective cohort; meter loads and secure subscriptions.
Rural space heat/home R_housing_space	14 [10–20] MWh/home/year useful	A · prospective cohort; meter loads and secure subscriptions.
Urban electricity U_electricity	225 [200–300] USD/MWh all-in	A · S22 p146 context only; obtain interval tariff quote.
Commercial avoided heat U_commercial_baseline	195 [160–230] USD/MWh useful	A · verify avoidable variable bills; retain fixed charges.
Public/housing avoided heat U_public_baseline	100 [80–130] USD/MWh useful	A · verify avoidable variable bills; retain fixed charges.
Urban pipe + civils U_pipe_usd_m	2500 [1500–4000] USD/trench m installed pair	A · screening cost; obtain scoped installed bids.
Rural pipe + civils R_pipe_usd_m	1100 [700–1800] USD/trench m installed pair	A · screening cost; obtain scoped installed bids.
Installed HP cost U_hp_usd_kw	600 [450–1000] USD/kW heating	A · screening cost; obtain scoped installed bids.
Urban grid carbon U_grid_ef	0.442 [0.2–0.6] tCO2e/MWh	A · S22 p147 illustration; verify accounting-year factor.
Gas boiler efficiency P_boiler_eff	0.85 [0.75–0.95] usefulheat/fuel HHV	A · explicit accounting allowance; validate local baseline / EPD.
Embodied allowance P_embodied	600 [250–1200] tCO2e urban installation	A · explicit accounting allowance; validate local baseline / EPD.

The full register additionally covers each installed CAPEX line, O&M, source royalty, fund, combined insurance/access allowance, profile shapes, discount rate, replacement and public/commercial discounts. Base prices are unverified contracts. S22's electricity estimate assumes a much steadier load than the modeled heat plant.

One numerical source of truth.

Edit inputs, rerun the model, regenerate browser data and rebuild the documents. Never edit a reported figure by hand.

Calculation	Definition / boundary
Source and heat pump	QHP = Qsource + Wcompressor. Heating COP = ηCarnot × Tcond/(Tcond − Tevap), temperatures in K. Tevap uses mean source water minus capture and evaporator approaches; Tcond uses supply plus condenser approach.
Distribution	Tout = Tground + (Tin − Tground) exp[−U L/(ṁ cp)] for each pipe. Total trench applies to both supply and return. Flow increases at low duty to preserve minimum delivery temperature; pressure includes both pipes and source/station allowance.
Storage / hourly dispatch	Usable energy = V ρ cp ΔT × usable fraction. Standing losses and finite energy and heat-pump charging limits included. Cyclic initial = final SOC; no free initial heat. Customer backup serves all unmet heat. Annual profiles are deterministic synthetic shapes.
Project cash flow	Revenue = metered delivered MWh × cohort tariff. OPEX includes all new electricity, fixed O&M/admin and the combined insurance/access placeholder, source royalty and fund. Year-15 HP refurbishment. NPV discounts 25-year unlevered pre-tax cash flow at7%; no terminal value.
Payback / LCOH	Simple payback uses undiscounted cash flows including refurbishment when relevant; discounted payback uses discounted flows. LCOH = discounted total project cost / discounted useful heat. No-payback results remain null.
Net operational carbon	Displaced baseline heat emissions minus compressor/pump/control/booster electricity emissions and refrigerant allowance. Retained backup is explicitly in whole-service totals. Embodied screening allowance is reported separately.
Distance lab	A steady-state required-duty solver, not dispatch. Tests available source, temperature, source capacity and HP capacity. Branch pipe CAPEX excludes shared plant and interfaces; variable cost excludes fixed overhead and sanitary boost.

Verification checks hourly heat conservation, capacity constraints, cyclic storage, backup, money/carbon reconciliation, scenario thresholds and Python–JavaScript parity. Model inputs are not statistically calibrated. The deterministic stress-grid pass fraction is not a real-world probability. Measured data and certified maps are required before engineering selection.

Known limits: no surveyed branched hydraulic network; no manufacturer part-load map; simplified tank stratification; no dynamic failure response; no optimized electricity tariff/dispatch; no observed building profiles; no site permits/contracts; no full lifecycle inventory. Reference year 2025 is an indexing convention, not measured weather.

Read in full.
Used with boundaries.

The supplied folder is the evidence corpus. Source IDs remain stable across the model, proposal, pitch and audit.

ID	Original file	Use / caution
S01	Decarbonization-and-Heat-Re-Use-in-Data-Centers.mp4	Full video transcript reviewed; heat-reuse context, not site telemetry.
S02	NYU_Hackathon_Data_Center_Heat_Reuse_Challenge_R1.docx.pdf	Governing challenge; all site and judging bullets.
S03	rubic1.png	Rubric image: quality criteria.
S04	rubic2.png	Rubric image: quality criteria.
S05	20230623 Data Centers HeatReuse 101 3.2.docx.pdf	pp4,6–10: energy, capture, hosts and reliability.
S06	district-energy-application-guide-district-energy-2018-master-en.pdf	pp7,13–15,23–25: isolation, flow, pressure, water quality.
S07	2026 10 01 Hackathon NYU - Suburban Site - Lake Hawkeye.pdf	Site context for proposed Lake Hawkeye; no measured thermal profile.
S08	2026 10 01 Hackathon NYU - Urban Site - 111 8th Ave.pdf	Urban local context; contradictory dashboard demographics excluded.
S09	NYU_Hackathon_Data_Center_Heat_Reuse_Challenge_R1.docx	Challenge Word original; equivalent scope to S02.
S10	iGRID Playbook.pdf	pp149–177: zones, measurement and supervisory optimization.
S11	IDEA - Thermal Energy Storage for Data Centers_CB and I.pdf	pp5,9–12: storage purpose and scope; no imported small-tank price.
S12	June 10th FPCJ Meeting Summary Compiled.pdf	pp5,18,20–22: retrofit example and asset-lifetime caution.
S13	Resource Efficient Decarbonization.pdf	Building-system retrofit precedents; not a local demand dataset.
S14	urbs Williams College Energy Transition Strategy ver1.0-komprimerad.pdf	p6: retained backup; campus transition context.
S15	NYU BAC Hackathon_HDR_Waste Heat Reuse_2026.1002.pdf	p42 generic capture temperatures; p17–18 metric cautions.

Read in full.
Used with boundaries.

The supplied folder is the evidence corpus. Source IDs remain stable across the model, proposal, pitch and audit.

ID	Original file	Use / caution
S16	NYU_Data_Center_Heat_Reuse_Hackathon_Deck_Challenge and Workshop- opening slides.pptx	Opening workshop; 13 actual slides plus notes.
S17	Topic 5 - Heat reuse, Connecting DC to DE systems_v5.pptx	s26 heat balance, s40 high-lift COP example; 44 slides plus notes and embedded video.
S18	CBS_Data_center_white_paper_DISTRICT HEATING.pdf	pp9–20: temperature, COP, distance and commercial screening.
S19	Colocating-Data-Centers_Greenhouses-RII-Virginia.pdf	pp36–41 greenhouse seasonality; pp24–26 continuity/community.
S20	Danfoss Waste Heat.pdf	Waste-heat context and technology examples; no site guarantee.
S21	Danfoss White Paper Sample.pdf	Recovery-system context; no local cost quote.
S22	DATA HEAT - Sector Couling Data Centeres and District Energy - Market Development Guide and Appendicies - 3MAR26.pdf	pp51–52 carbon, p77 tenor, pp146–147 indicative NYC tariff/grid.
S23	Developing Community and Data Center Synergies With District Energy - CenTrio.pdf	pp3–7,15: community needs, finance and engagement.
S24	driving_sustainable_data_centres-AD Highlighted.pdf	pp11,21,23: redundancy, leases and KPI boundaries.
S25	Every Drop Counts - Grundfos Water Scarcity Paper_2026_Web (1).pdf	pp10–13: water boundary; no site water-saving quantity.
S26	US Policy Landscape Data Center Heat Reuse - David Gardener & Associates.pdf	Policy landscape; no grant booked from proposed legislation.
S27	Williams_College_and_the_Geothermal_Earth_Battery.m4a	Complete audio reviewed; broad resilience claims checked against S14.
S28	The_Hacker_Toolkit_TEMPLATE_AND_EXAMPLES_ONLY.pdf	pp7,9,11: Canvas, CPR and pitch structure only.
S29	Discord Message Accouncement (new).txt	Software and explainable confidence scope clarification.

Reading included all extracted source text, selected diagram/image-only pages, complete video/audio transcripts and the embedded Mainz video. The reading ledger records visual coverage explicitly. Duplicate workshop notes and generic examples were not treated as site observations.

Source conflicts excluded: context demographic caption/dashboard discrepancies; generic short-payback claims without local cash flow; incorrect/ambiguous ERE/ERF statements; greenhouse recovery limitations presented as universal heat availability; proposed policy incentives presented as available funding. Original files were not altered.